Data Visualization & Analytics Lab

Experiment No. : 08

Develop a time series model

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1. **Aim:** Develop a time series model on a given dataset and visualize trends over time..

2. **Objectives:**
   - Understand why visualization is an important part of time-series analysis.
   - Understand the components of time-series data.

3. **Course Outcomes**: Students will be able to understand the challenges and methods for visualizing time-series data.

4. **Hardware / Software Required:** Python/Tableau

5. **Theory:**
   Time series analysis is a specific type of data analysis that focuses on data points that are ordered by time, where time is the independent variable.

   The primary goal is to understand the underlying patterns and relationships within the data to make predictions or forecasts about future values.

   Key Components:

   Trend: The overall direction or movement of the data over time (e.g., increasing, decreasing, or stable).

   Seasonality: Periodic fluctuations or patterns that occur at regular intervals (e.g., annual, quarterly, monthly).

   Cyclicity: Longer-term fluctuations that don't have a fixed period like seasonality (e.g., economic cycles).

   Irregularity: Random or unpredictable fluctuations that cannot be attributed to the other components.

Implementation of Time-Series analysis in Tableau

To perform time series analysis with **smoothing** in Tableau, you can leverage Tableau's built-in forecasting capabilities, specifically the exponential smoothing model, which helps identify patterns and forecast future values by giving more weight to recent data points.

Steps:

1. Prepare Your Data:

- **Connect to your data source:**

  Ensure your data includes a date/time dimension and the measure you want to analyze (e.g., sales, website traffic).

- **Structure your data:**

  Make sure the date/time field is recognized as a date/time field in Tableau.

- **Aggregate data (if needed):**
  Choose the level of granularity (e.g., yearly, monthly, daily) that best suits your analysis.

2. Create a Time Series Chart:

- **Drag and drop:**

  Drag your date/time field to the "Columns" shelf and your measure to the "Rows" shelf.

- **Choose a visualization:**

  Tableau will typically default to a line chart, which is ideal for time series data.

- **Drill down:**

  Use the plus icon on the date field to explore different levels of granularity (e.g., years, months, days).

- **Make it continuous:**
  Right-click on the date field and select "Continuous" to create a continuous time series chart.

3. Add Forecasting:

- **Right-click and select "Show Forecast":** Right-click on the view and select "Show Forecast" to add a forecast line.
- **Explore forecast options:** Tableau will automatically apply an exponential smoothing model, but you can customize the forecast options:

- **"Actual & Forecast":** Shows both actual data and the forecast.
- **"Trend":** Displays the underlying trend line.
- **"Precision":** Shows the confidence intervals for the forecast.
- **"Precision %":** Shows the confidence intervals as percentages.

- **Evaluate the model:** Tableau provides metrics like Alpha, Beta, and Gamma smoothing coefficients, which can help you understand the model's performance.

4. Understand Exponential Smoothing:

- **Weighted average:**

  Exponential smoothing uses a weighted average of past values, with more recent values having greater weight.

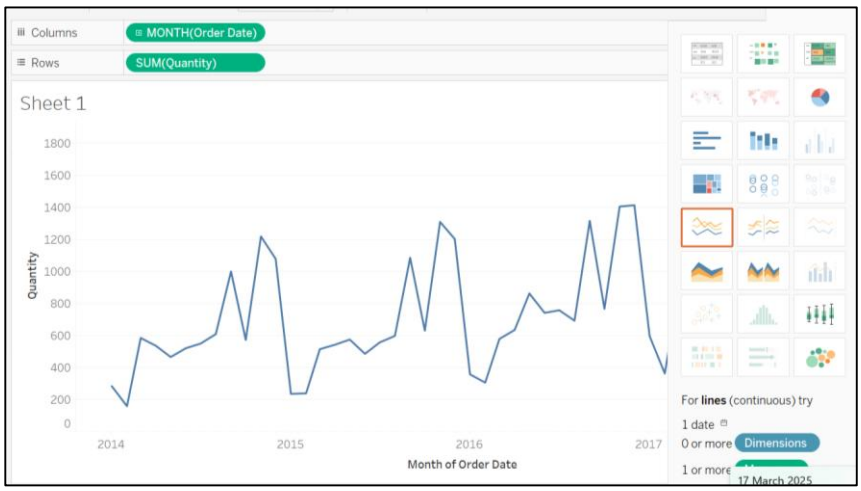
- **Holt-Winters method:**

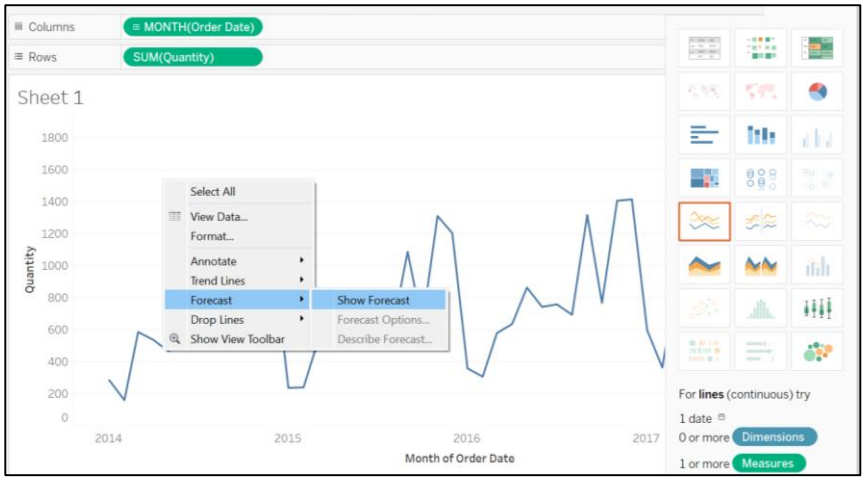
  Tableau uses a variant of the Holt-Winters method, which is an exponential smoothing technique designed to handle seasonality.

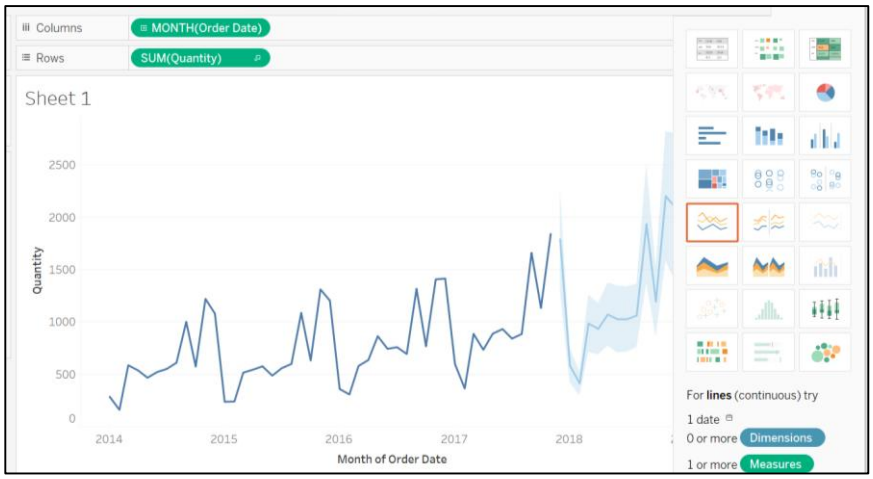
- **Trend and Seasonality:**

  The model can capture both trend and seasonal components in your data.

- **Forecast Algorithm:**
  The forecast algorithm tries to find a regular pattern in your data that can be continued into the future.







Smoothed Data Visualization:

- **Create a Calculated Field:**
- **Moving Average:** Use the `MOVING_AVG()` function to calculate a moving average of your measure.

  OR

  Right-click on the field you want to use for the moving average (e.g., "SUM(Sales)") and select "Quick Table Calculation".

  Choose "Moving Average" from the Quick Table Calculation menu.
- **Exponential Smoothing:** Tableau offers built-in forecasting capabilities using exponential smoothing models.

Implementation of Time-Series analysis in Python

Steps:

1. Import required Python libraries.
2. Load the time-series dataset.
3. Convert the date column into datetime format.
4. Set the date column as the index.
5. Plot the time-series data.
6. Calculate rolling statistics (mean and standard deviation).
7. Visualize trends and patterns.
8. Interpret the results.

Sample code:

```
# Import required libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load dataset
data = pd.read_csv("timeseries_data.csv")

# Convert date column to datetime
data['Date'] = pd.to_datetime(data['Date'])

# Set date as index
data.set_index('Date', inplace=True)

# Display first few rows
print(data.head())

# Plot time series
plt.figure(figsize=(10,5))
plt.plot(data['Value'])
plt.title("Time Series Data Visualization")
plt.xlabel("Date")
plt.ylabel("Value")
plt.show()

# Rolling statistics
```

```
rolling_mean = data['Value'].rolling(window=12).mean()
rolling_std = data['Value'].rolling(window=12).std()

# Plot rolling statistics
plt.figure(figsize=(10,5))
plt.plot(data['Value'], label='Original')
plt.plot(rolling_mean, label='Rolling Mean')
plt.plot(rolling_std, label='Rolling Std')
plt.legend()
plt.title("Rolling Statistics")
plt.show()
```

6. Conclusion:

We have implemented time-series data analysis model with visualization to display the trends. We also performed forcasting. The smoothing techniques in time-series analysis is used to provide average forecasting.

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7. Viva Questions:

- What is time-series data? What are the components of it?
- What is the importance of smoothing in time series analysis?

References:

1. https://www.rigordatasolutions.com/post/tableau-moving-average
2. https://community.tableau.com/s/question/0D54T00000CWdXmSAL/moving-averages-with-edge-smoothing-by-triangular-windows